

Caleb Brown

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Education

Auburn University – BS in Software Engineering

Aug 2022 – Summer 2027

Technical Skills

- **Languages & Frameworks:** Go/Gin, Swift, C#, Zig, JavaScript/TypeScript, React, SQL/PostgreSQL, Unity
- **Cloud & DevOps:** Microsoft Azure, Kubernetes, ArgoCD, CI/CD, GitHub Actions, Git
- **Hardware & Standards:** RAIN RFID, NFC, 2D Barcode Systems, Zebra Handheld Readers, GS1
- **Engineering & Leadership:** Systems Architecture, Microservice Architecture, Code Review, Project Management, Team Leadership, Agile & Scrum, Product Lifecycle Management

Relevant Experience

Product Development Manager, Auburn University RFID Lab – Auburn, AL

Aug 2023 – Present

- Built and maintained an RFID tag/inlay performance auditing system that audited 500K+ individual items for ARC and ALEC compliance across 12+ Fortune 500 partners in the US, Canada, and Mexico using React, Go/Gin, and Swift (iOS App)
- Owned the full software development lifecycle for 15 services and applications, setting coding standards, running code reviews before release, and building an automated test suite covering performance and error handling
- Screened 20+ candidates and hired 9 engineers; onboarded and mentored the team with feedback loops and development plans, maintaining 80% team retention
- Maintained DevOps infrastructure on Azure across 3 Kubernetes clusters, automating deployments with ArgoCD and GitHub Actions CI/CD pipelines and enabling zero-downtime rollouts with at least two replicas per service
- Built an iOS app alongside GS1 engineers to support a global 2D barcode adoption study across GS1's 100+ member countries, delivering partners data tools and infrastructure

Training Director, Chick-fil-A – LaGrange, GA & Auburn, AL

Aug 2020 – Aug 2025

- Directed shift operations for a \$12M restaurant, coordinating up to 40 staff members per shift
- Managed peak service periods, sustaining 150+ cars and \$2,000+ in sales per hour through updated staffing and workflows
- Conducted daily mock health inspections, maintaining 98+ health inspection scores and 99%+ corporate audits
- Designed and led training programs that raised service skill, product knowledge, and operational efficiency
- Coached trainees with targeted feedback, developing multiple team members promoted to trainer and leadership roles

Relevant Projects

EAGLE Platform

eagle.rfidlab.org

- Designed a multi-scope platform with a microservice architecture on Azure Kubernetes for RFID performance auditing in partner warehouses and stores across the US, Canada, and Mexico, and ALEC item submissions from 5,000+ vendors
- Ensured reliability of services through separation of responsibility, allowing multi-service failover without customer impact, using separate backends and frontends as well as isolated authentication and authorization
- Built a dedicated iOS mobile application that integrates with Zebra handheld RFID readers by writing a Swift integration SDK from scratch, automating data collection and enabling lab teams and partners to audit their own items for the first time
- Grew the effort to a team of 10 software engineers developing and maintaining new features for 12+ Fortune 500 partners while supporting customer service and quality assurance

RFID Trilateration & Sensor Fusion

1st Place, Auburn RFID Lab Hackathon

- Built RFID trilateration with three antennas and a single Zebra FX9600, using a Zig backend with PostgreSQL to calculate RSSI positioning within a 10 ft radius at 1–2 second updates
- Built a LiDAR and RFID sensor fusion demo that located 6 RFID-tagged items on a shelf in under a second using unique identifiers stored in PostgreSQL

Involvement

President, Chi Phi Fraternity – Auburn, AL

Nov 2024 – Nov 2025

- Led a nine-member executive council for a 30+ member chapter, growing annual recruitment 63% (8 to 13 new members)
- Served as the primary point of contact with Auburn University and external partners, keeping the chapter in good standing, securing event approvals, and organizing a New Orleans formal

Director of New Member Experience, Auburn University Interfraternity Council – Auburn, AL

Nov 2023 – Nov 2024

- Collaborated with all 28 Auburn University fraternities to coordinate and standardize education for 1,000+ new members
- Organized “Auburn Man,” a speaker-led event series dedicated to inspiring and educating the next class of fraternity men